

Executive Summary:

Scientific Concept Diffusion Across Disciplines

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Headline Finding

Concepts whose early cooccurrence network is topically diverse — connecting to partners from different research communities — spread to more disciplines later. The $\text{OPEN}_{\text{home}}$ index, a six-component measure of ego-network openness built on home-field cooccurrences only, has a partial Spearman priority of **+0.069** [+0.038, +0.100] with rarefied cross-field breadth, pooled over six non-selection bodies ($I^2 = 0$, 6/6 positive). The signal is topical non-redundancy rather than temporal partner turnover.

Data and Design

The study used the OpenAlex bulk snapshot (476M works) to build a panel of 12,499 scientific concepts (onsets 2003–2014) spanning 27,393 concept-by-field adoption episodes across 26 venue fields. The primary outcome was rarefied field breadth (O_{2r} , $m = 50$): the expected number of distinct venue fields among a fixed-size random draw of papers from the outcome window ($t_0 + 6$ to $t_0 + 8$). All indicators were computed over the early window t_0 to $t_0 + 2$.

The run used a hash-sealed pre-registration protocol with a single-unseal design. DEV concepts (CS/Eng/BGM/Med homes, 2003–2009 onsets, $n = 4,771$) were used for indicator selection. Held-out groups (Physical Sciences, Life & Environment, Social Sciences, Math & Decision Sciences, $n = 3,372$) and an onset cohort (2010–2014, $n = 4,356$) were scored once. A fresh 2015–2017 cohort ($n = 1,443$) served as independent confirmation.

Key Results

- Seven of 53 early indicators predict cross-field breadth on held-out fields.** Four belong to the retained-frontier family (M0.density_end, D.vol_end, CONTACT_REACH, RETENTION_RATIO) and three to cooccurrence topology (n_{comm} W3, NOV_{res}, ego_density W3). All seven had 6/6 sign agreement across evaluation units.
- Consistency predicts volume but narrower reach.** Cheng et al.’s ideational consistency predicted next-year volume at +53.5%/SD in their design, but only +1.3% with current-size control. As an early trait, consistency predicted *narrower* cross-field reach: partial Spearman -0.069 (DL -0.079 , $I^2 = 0$, 5/5 negative).
- Concepts spread to fields related to those that retain them.** On an independent 11,841-concept frame, the conditional-logit coefficient for retained-field relatedness was $d_0 = 0.322$ [0.291, 0.355] (LR = 325.8). The volume-matched contrast was null (verdict: PARTIAL).

4. **Breadth is 73% early exploration, 27% retention.** The decomposition of the top-vs-bottom breadth tercile gap showed early contact diversity contributed 78%, frontier advance –5%, and retention 27%. Broad concepts start with wider contact, not by advancing a wider frontier.
5. **New-community partners from mixed-field papers carry the signal.** Partners from new Leiden communities ($C_2 = +0.102$) and mixed-field papers ($C_4 = +0.103$) carry the openness–breadth association. Domain novelty, not methodological novelty, is the main contributor.

Dead Ends

- The naturalisation gap A_h^* failed every pre-registered clause (iteration 1).
- Gateway centrality predicting field retention was disconfirmed at scale: $\Delta\text{AUC} = -0.00001$ (iteration 2).
- About half of the top two indicator effects (M0_density_end, D_vol_end) was pre-onset footprint (attenuation 0.50 and 0.45).
- Within-concept closure (Exp11/15) was NOT SUPPORTED on DEV and opposite-signed on held-out.
- Two artifacts failed outright (Exp2, Exp9) and one was incomplete until iteration 5 (Exp11).

Limitations and Open Questions

The OPEN–breadth association is modest and adds no practical forecasting gain over the popularity baseline (+0.002 Spearman). Split-half reliability is 0.49. The within-concept permutation null absorbed temporal churn, but the excess-churn variants were too unreliable (SB ~ 0.01 –0.05) to adjudicate whether temporal turnover matters. Turnover is *untested*, not refuted. The retained-field relatedness effect reverses under minimum-conditional-probability proximity and shows high heterogeneity ($I^2 = 0.92$). The Cheng reversal was not confirmed on Frame N. No causal mechanism was established.

Evidence State

Lead / exploratory. The paper headline is **OPEN_{home} PSP +0.069 [+0.038, +0.100]**, pooled over 6 non-selection bodies, $I^2 = 0$, 6/6 positive. Five iterations, 20 artifacts commissioned, 16 completed. All review scores were blocking (3/3/3/2/2).